

THUNDERBOLT.AI

Local Large Language Model Benchmark

Performance, Quality and Hardware Efficiency Evaluation

BENCHMARK STATUS Completed evaluation	MODELS TESTED 3 local language models
TEST QUESTIONS 10 standardized questions	INFERENCE Local Ollama runtime

Prepared from measured local inference results

1. Executive Summary

Thunderbolt.ai was evaluated as a local large language model benchmarking system using three models running through Ollama on a laptop equipped with an Intel Core i7-13620H processor and an NVIDIA GeForce RTX 4050 Laptop GPU with 6 GB of VRAM.

The evaluation used a common set of ten questions covering coding, debugging, reasoning and technical knowledge. For each request, the system recorded response latency, generated tokens, generation speed and hardware utilization.

The measured results produced **Gemma 3 4B** as the highest-scoring model under the weighting scheme used by this benchmark. The result is specific to the tested workload and hardware configuration and is not intended to represent a universal ranking of the evaluated models.

Gemma 3 4B

Overall Winner

Gemma 3 4B

Best Quality

Gemma 3 4B

Fastest

2. Evaluation Environment

Component	Measured Configuration
Processor	13th Gen Intel Core i7-13620H
Graphics	NVIDIA GeForce RTX 4050 Laptop GPU
GPU Memory	6 GB
Inference Runtime	Ollama
Execution	Local inference
Models	Qwen2.5-Coder 7B, Gemma 3 4B, Qwen3 8B
Benchmark Questions	10

3. Models Evaluated

Model	Parameter Class	Primary Evaluation Role
Qwen2.5-Coder 7B	7B	Coding and debugging
Gemma 3 4B	4B	General-purpose local inference
Qwen3 8B	8B	General reasoning and technical tasks

4. Benchmark Methodology

All three models were evaluated using the same benchmark questions and the same local inference environment. This provides a controlled basis for comparing their measured performance.

The benchmark categories were coding, debugging, reasoning and technical knowledge. Each model received ten questions through the same benchmark runner.

Measured inference metrics:

Metric	Purpose
Response time	Measures total request latency.
Generated tokens	Measures response length.
Tokens per second	Measures generation throughput.
CPU utilization	Measures processor load.
RAM usage	Measures system memory consumption.
GPU utilization	Measures GPU workload.
VRAM usage	Measures GPU memory consumption.

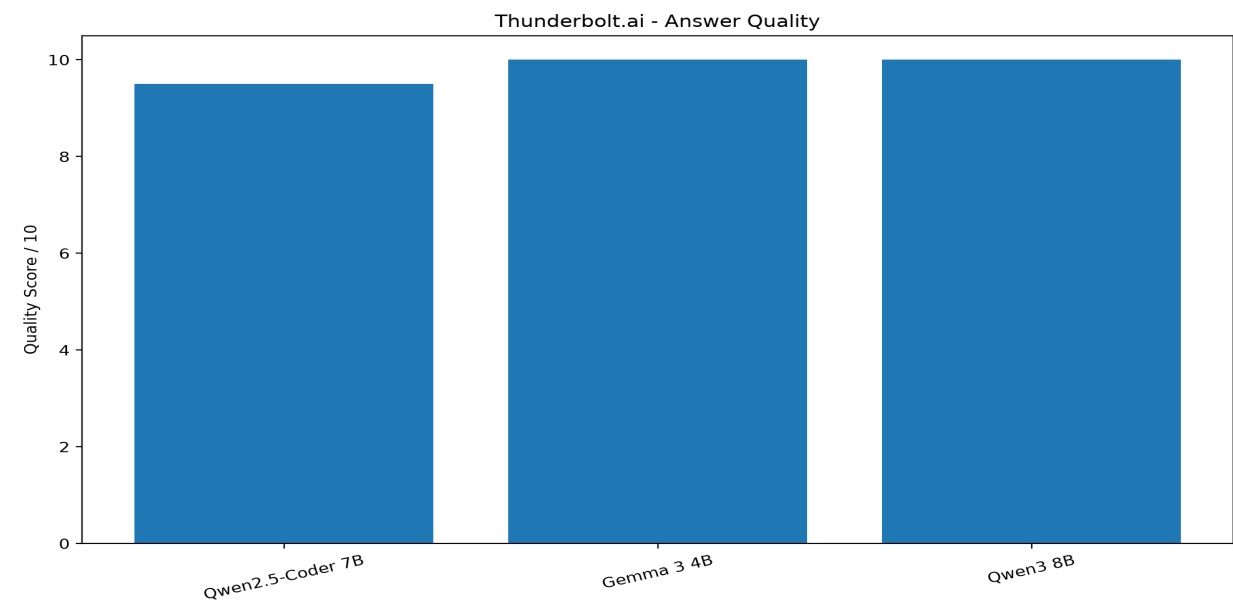
5. Measured Results

Rank	Model	Quality	Time	Tok/s	VRAM	Score
1	Gemma 3 4B	10.0	23.93s	54.70	3.78 GB	97.31
2	Qwen2.5-Coder 7B	9.5	29.84s	17.58	4.14 GB	73.65
3	Qwen3 8B	10.0	171.90s	12.60	4.34 GB	65.25

6. Quality Evaluation

The quality evaluation assigns a score to each benchmark response according to predefined criteria associated with the individual question. The resulting scores represent performance on this benchmark set rather than a universal measure of model intelligence.

Figure 1. Quality Score Comparison



7. Inference Speed and Latency

Generation speed is reported as tokens generated per second. Response time represents the total elapsed time for each benchmark request.

Figure 2. Generation Throughput

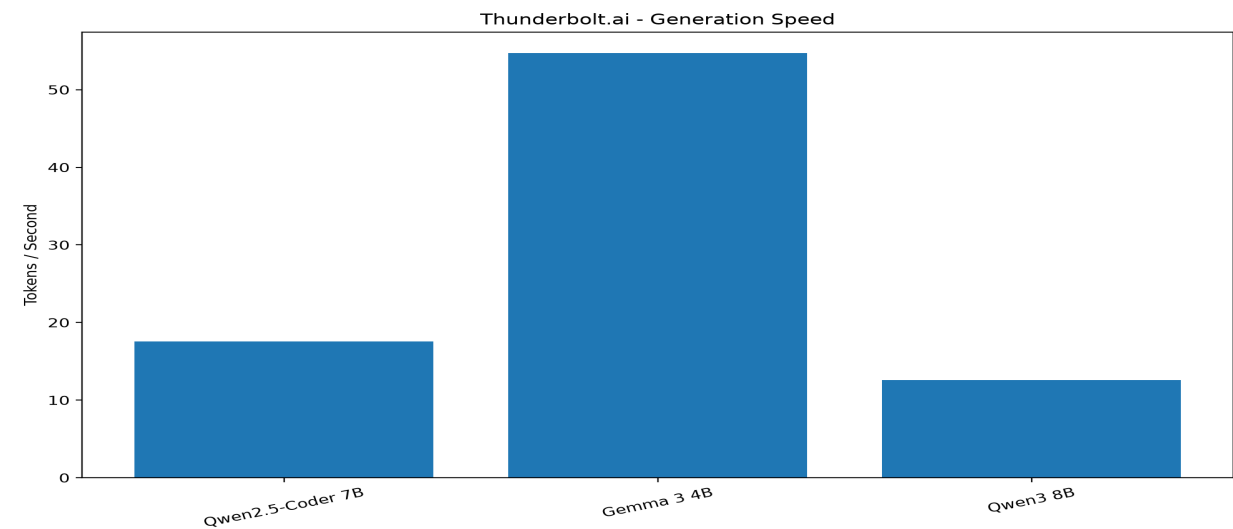
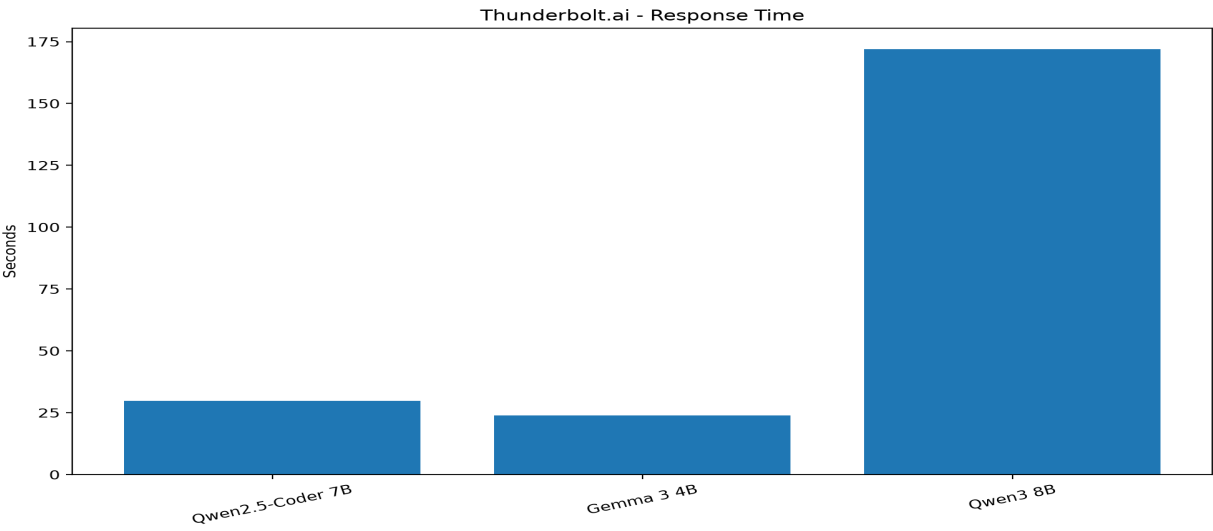


Figure 3. Average Response Time



8. Hardware Utilization

Hardware measurements show how heavily each model used the available CPU, system memory, GPU and VRAM during inference. These measurements are particularly relevant for local deployment because the laptop has a finite 6 GB VRAM budget.

Figure 4. Average CPU Utilization

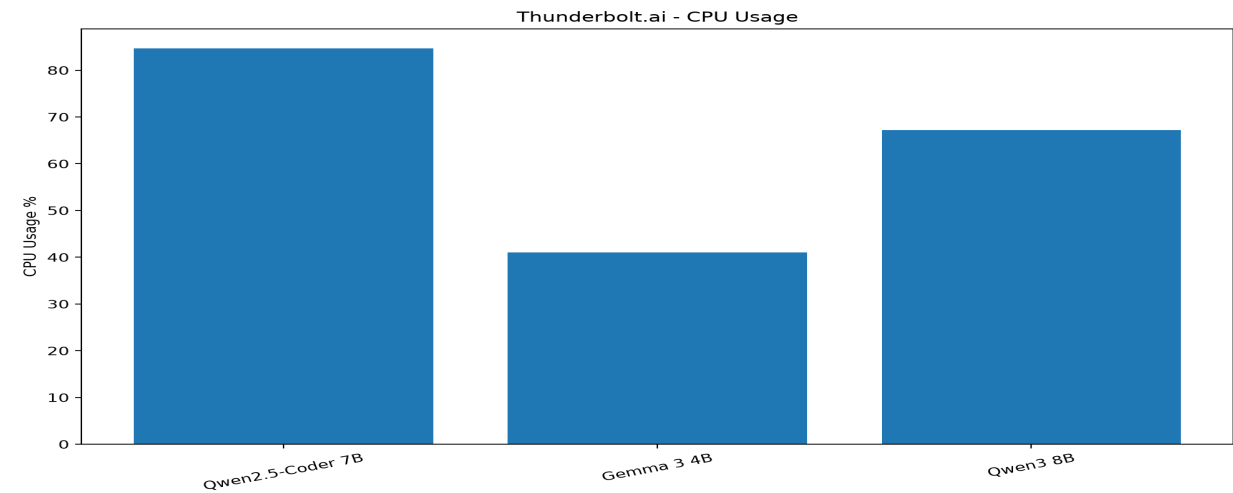


Figure 5. Average RAM Usage

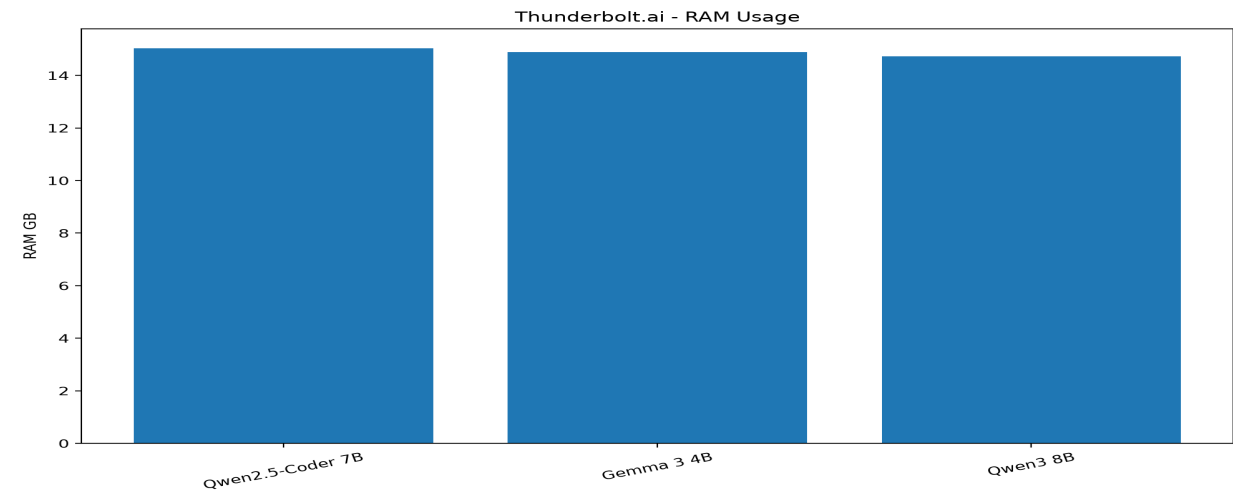


Figure 6. Average GPU Utilization

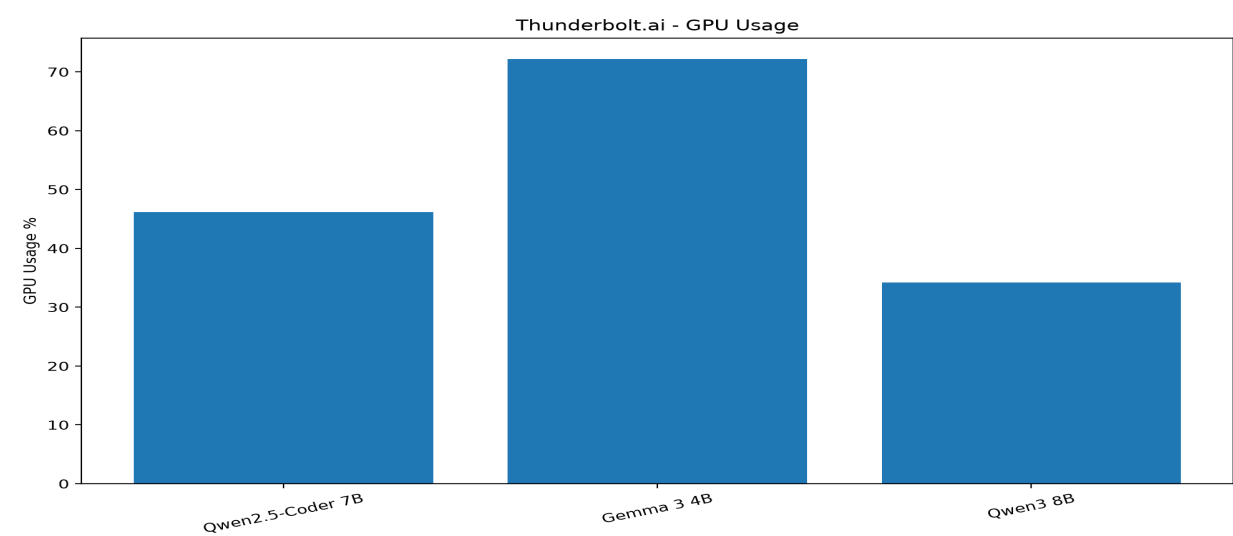
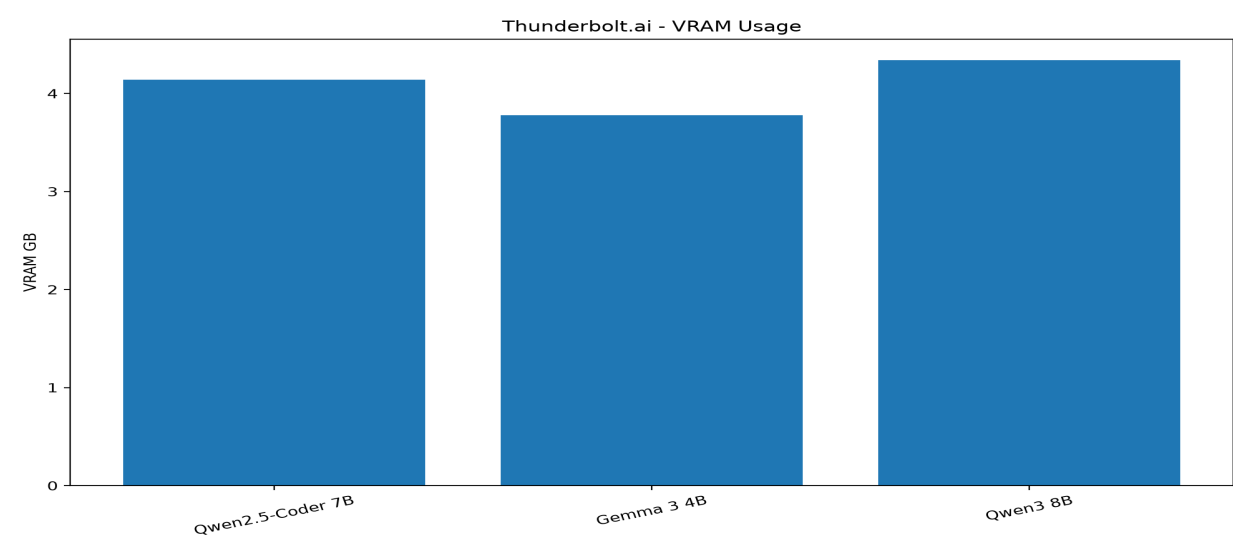


Figure 7. Average VRAM Usage



9. Final Analysis

Gemma 3 4B achieved the highest overall benchmark score under the weighting scheme implemented by Thunderbolt.ai.

The measured results show that Gemma 3 4B achieved a quality score of 10.00/10, an average generation speed of 54.70 tokens per second, an average response time of 23.93 seconds and an average VRAM usage of approximately 3.78 GB.

Qwen2.5-Coder 7B achieved a quality score of 9.50/10 and provided a coding-oriented alternative, while Qwen3 8B provided the largest parameter class among the evaluated models but produced substantially lower generation throughput on the tested hardware.

These findings demonstrate the importance of evaluating models using both output quality and system-level performance. Parameter count alone does not determine the most suitable model for a local deployment.

Benchmark Scoring Model

Component	Weight
Quality	40%
Generation speed	25%
Response latency	15%
Hardware efficiency	20%
Total	100%

10. Conclusion

The benchmark successfully established a repeatable local evaluation pipeline for comparing language models under a controlled hardware environment.

For the tested workload, Gemma 3 4B provided the strongest overall balance between answer quality, generation speed, response latency and hardware requirements.

Qwen2.5-Coder 7B remains a relevant option for coding-focused workloads, while Qwen3 8B demonstrates the performance cost associated with running a larger model on hardware with a 6 GB VRAM constraint.

The results documented in this report are measurements from the specified benchmark environment and should be interpreted as an engineering comparison of local inference performance.

Benchmark Record

Models evaluated	3
Questions per model	10
Evaluation categories	Coding, Debugging, Reasoning, Technical

Inference runtime	Ollama
GPU memory available	6 GB
Report basis	Measured local inference results